SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE

Department of Computer Science Engineering

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Lab Objectives

- Understand Test Driven Development (TDD)
- Write test cases before implementation
- Validate functionality using assertions
- Improve code reliability using testing practices

Task 1: Basic Function Testing

Python Code

```
def add(a, b):
    return a + b

assert add(2, 3) == 5
assert add(-1, 1) == 0
assert add(0, 0) == 0
```

Explanation: Simple function tested using assert statements.

Task 2: Prime Number Testing

Python Code

```
def is_prime(n):
    if n <= 1:
        return False
    for i in range(2, int(n**0.5)+1):
        if n % i == 0:
            return False
    return True

assert is_prime(2) == True
assert is_prime(4) == False
assert is_prime(7) == True
```

Explanation: Function is validated for multiple test cases.

Task 3: String Reverse Testing

Python Code

```
def reverse_string(s):
    return s[::-1]

assert reverse_string("abc") == "cba"
assert reverse_string("hello") == "olleh"
assert reverse_string("") == ""
```

Explanation: Tests confirm correct string reversal.

Task 4: Exception Testing

Python Code

```
def divide(a, b):
    return a / b

try:
    divide(5, 0)
except ZeroDivisionError:
    print("Division by zero handled")
```

Explanation: Exception handling ensures safe execution.

Task 5: TDD Workflow

```
# Step 1: Write test
assert multiply(2, 3) == 6

# Step 2: Write function
def multiply(a, b):
    return a * b

# Step 3: Run tests
assert multiply(2, 3) == 6
assert multiply(0, 5) == 0
```

Explanation: TDD follows: Write Test → Implement → Validate.

Conclusion

Test Driven Development ensures correctness and reliability.

Writing tests first helps detect errors early and improves code quality.